Perceptual Priority Computing (PPC):

A Subject-First Architecture for Progressive Image Generation on CPU-Constrained Hardware

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ABSTRACT

Current image-generation systems allocate equal computational effort to every pixel simultaneously, requiring the entire scene to complete before the user receives any visual feedback. This paper introduces **Perceptual Priority Computing (PPC)** and its primary realization, **Saliency-Weighted Progressive Diffusion (SWPD)** — a staged generation architecture that decomposes a text prompt into a perceptual hierarchy and generates visual elements in order of human attention priority. The primary subject is generated first using a lightweight single-step diffusion model and displayed to the user immediately. The background environment is then generated as a separate conditioned inpainting pass, built around the subject's pose, lighting, and spatial position, and progressively revealed as generation completes. On CPU-only hardware, this approach reduces *perceived latency* by approximately 10× without increasing total computation. The paper also proposes PPC as a general computational philosophy with promising extensions to language reasoning, video generation, and other computationally intensive AI tasks.

Keywords: *perceptual priority computing, image generation, diffusion models, progressive rendering, CPU inference, inpainting, saliency, perceived latency, staged generation*

1. Introduction

Modern text-to-image generation systems — Stable Diffusion, DALL-E, Midjourney, and their successors — treat image synthesis as a single monolithic computation. From the first denoising step, the model simultaneously constructs every element of the scene: subject, background, shadows, textures, and atmospheric details. While this produces high-quality results, it carries a fundamental user-experience limitation: no meaningful visual output is delivered until the entire image is complete.

On GPU hardware this wait is typically only a few seconds. On CPU hardware — the primary compute resource for the majority of personal computers worldwide — the delay extends to several minutes. During this time the user sees nothing and has no sense of progress. This represents a practical barrier to democratizing image generation beyond the GPU-equipped minority.

This paper proposes a different philosophy, termed **Perceptual Priority Computing (PPC)**. The central observation is simple: *generate what matters first, and generate what matters less later*. Human visual attention is not uniform — people fixate on the primary subject before inspecting background, texture, and atmospheric detail. Human artists have always worked this way: sketch the subject first, build the environment around it. PPC formalizes this centuries-old observation as a computational architecture for AI image generation.

1.1 Motivating Example

Consider the prompt: *"A cat sitting in a garden."* A traditional pipeline begins constructing cat, grass, sky, shadows, and flowers simultaneously; the user waits for the full scene. A PPC-based system instead: (1) identifies *cat* as the primary perceptual subject; (2) generates the cat using a lightweight fast model and displays it immediately; (3) generates the garden environment conditioned on the existing cat; and (4) progressively reveals the background from blurred to sharp while the subject stays crisp — finally producing a complete, coherent image.

1.2 Contributions

The main contributions of this paper are:

(1) The Perceptual Priority Computing (PPC) framework — a new computational philosophy for visual generation.

(2) Saliency-Weighted Progressive Diffusion (SWPD) — a concrete four-stage architecture implementing PPC.

(3) Subject-conditioned background generation that preserves lighting, scale, and perspective coherence.

(4) A progressive compositor that keeps the subject sharp while the background reveals gradually.

(5) A working CPU-only implementation using open-source diffusion models with detailed timing results.

(6) A quantitative analysis of perceived latency reduction versus actual compute cost.

(7) A theoretical extension of PPC to language reasoning as a future research direction.

2. Core Principle: Perceptual Priority Computing

Perceptual Priority Computing (PPC) rests on a well-established finding in visual cognition: human gaze is strongly drawn to subjects and objects before backgrounds and fine detail [1]. When observing a scene, observers fixate first on the most salient object — its face, form, or implied motion — before exploring the environment. Modern image-generation systems ignore this asymmetry; they allocate computational effort uniformly across the spatial canvas regardless of semantic importance. The result is that resources constructing a distant tree or a sky gradient compete equally with resources constructing the subject's face.

PPC proposes a re-allocation: *computation should follow perceptual importance, not spatial position.* The system identifies which visual elements contribute most to a human observer's understanding and prioritizes those elements first — both in generation order and in the immediate feedback delivered to the user.

"Computational effort should follow perceptual importance."

2.1 Perceptual Hierarchy

Every visual prompt can be decomposed into a three-tier perceptual hierarchy. The PPC framework generates these tiers in strict priority order, displaying each tier as it completes rather than waiting for all tiers to finish:

Priority Tier	Visual Elements	Example: "tiger in a snowy forest"
Primary subject	Main noun, focal object	Tiger
Secondary elements	Environment, supporting context	Snow, trees, ground
Tertiary elements	Atmosphere, lighting, fine detail	Mist, shadows, distant trees

Table 1. PPC perceptual hierarchy: three tiers generated in priority order.

3. Related Work

3.1 Diffusion Models for Image Generation

Denoising Diffusion Probabilistic Models (DDPM) [2] established the foundation of modern text-to-image synthesis. Latent Diffusion Models (LDM) [5] reduced computational cost by operating in a compressed latent space, enabling practical high-resolution generation. SD-Turbo [6] further reduced inference to a single denoising step using adversarial diffusion distillation, making CPU inference feasible for subject-only generation — a key enabler of the SWPD pipeline.

3.2 Inpainting and Conditional Generation

Stable Diffusion Inpainting enables generating content within a masked region conditioned on the surrounding image context. This technique is central to SWPD Stage 3: the already-generated subject serves as surrounding context, and the background region is the masked area to be filled coherently. Recent work such as CAT-Diffusion [22] improves text-guided object inpainting by predicting semantic features of the target conditioned on unmasked context before the pixel-level generation pass — a conceptually adjacent idea to SWPD's two-pass structure.

3.3 Foreground-Conditioned Background Generation

The sub-problem of generating a coherent background around an existing foreground subject has received growing attention. The **Anywhere** framework [23] addresses foreground-conditioned image generation using a multi-agent pipeline that separates foreground understanding, diversity enhancement, and object integrity protection. It identifies foreground-background inconsistency and limited diversity as the core failure modes of prior end-to-end inpainting models. The **ATA** (Adaptive Transformation Agent) model, published at CVPR 2025 [25], further advances subject-position-variable background inpainting, adaptively determining a harmonious subject location and generating a coherent scene around it.

SWPD's Stage 3 is related to this line of work in that it also generates a background conditioned on a pre-existing subject image. However, the goals differ fundamentally: Anywhere and ATA target *image quality and spatial coherence*, whereas SWPD targets *perceived latency reduction* in a CPU-constrained user-facing pipeline. Neither prior work frames subject-first generation as a latency strategy or produces an intermediate user-visible output before the background is complete.

3.4 Saliency-Guided Image Generation

GazeFusion [10], published in ACM Transactions on Applied Perception (2024), presents a saliency-guided framework that conditions a diffusion model on a user-specified viewer attention distribution, causing generated images to attract viewer gaze toward desired regions. GazeFusion applies saliency as a *generation target* — controlling where viewers look in the output image. SWPD applies saliency as a *generation schedule* — controlling the computation order so that high-salience content is delivered to the user earliest. These are complementary but distinct uses of the same perceptual insight.

3.5 Progressive Image Rendering

Progressive JPEG encoding [8] established the principle of displaying a degraded version of the full image before detail loads — the whole image appears blurry initially and sharpens over time. SWPD differs fundamentally: the primary subject is rendered at full quality immediately — not as a blurry approximation of a complete scene. Progressive degradation in SWPD is applied only to the background, never the subject. This distinction means the user receives high-quality, semantically meaningful content at first display rather than a uniformly blurred placeholder.

3.6 Gap in Existing Work

To our knowledge, no prior work combines prompt decomposition into a perceptual hierarchy, subject-first diffusion with immediate user display, subject-conditioned background inpainting, and a progressive subject-anchored compositor into a unified CPU-targeted pipeline designed explicitly to reduce *perceived* latency. This orchestration — and the explicit user-experience framing of the problem — is the primary contribution of this paper.

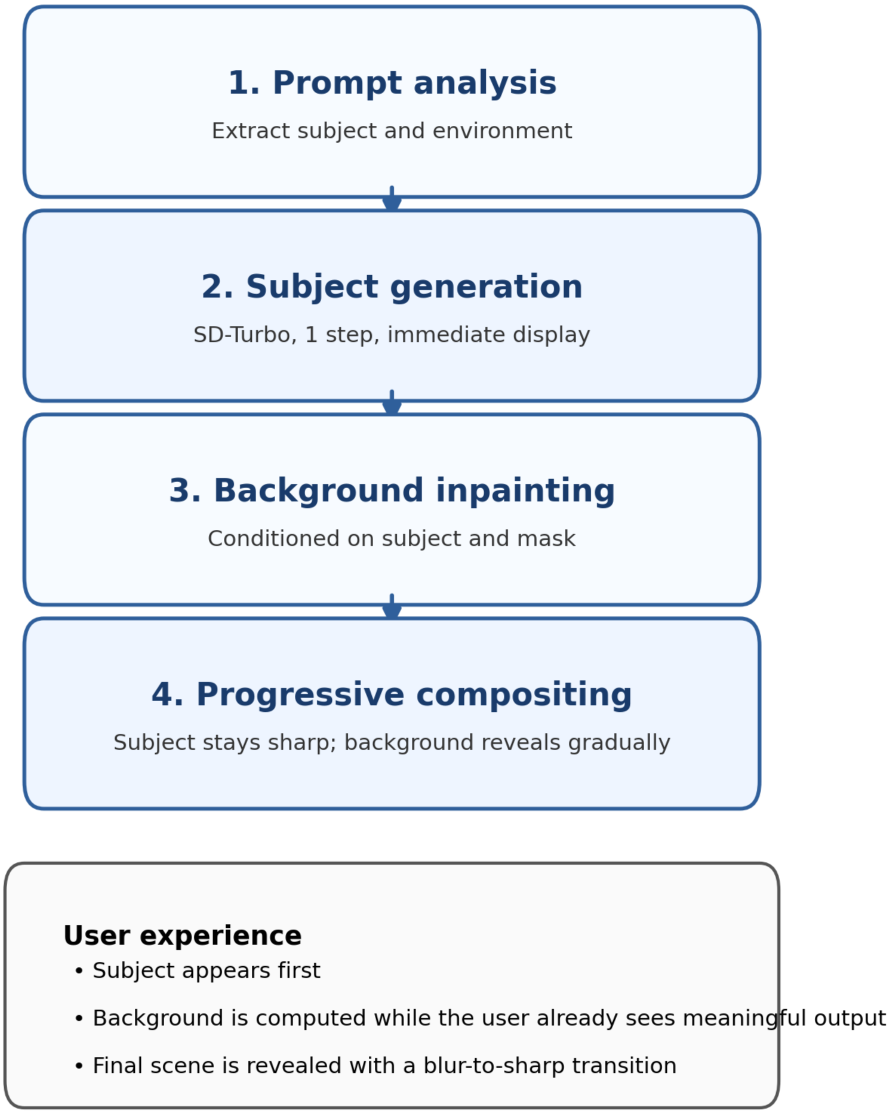
4. Proposed Architecture: SWPD

The Saliency-Weighted Progressive Diffusion (SWPD) architecture implements PPC through four sequential stages. Each stage builds on the output of the previous stage. Stages 2 and 4 both produce user-visible output at distinct quality levels.

Stage	Name	Output to User?	Key Operation

Figure 1. Saliency-Weighted Progressive Diffusion (SWPD) pipeline

Subject-first generation reduces perceived latency on CPU-only hardware.



Original contribution: a user-facing orchestration strategy that prioritizes perceptual importance in generation order.

1	Prompt Decomposition	No	Extract subject S and environment E; build P_subject, P_background
2	**Subject Generation**	**YES — immediately**	SD-Turbo (1 step) → subject at 512×512 on white bg
3	Background Generation	No	SD-Inpainting (25 steps) conditioned on subject image + mask
4	Progressive Compositing	YES — 12 frames	Blur-to-sharp background reveal; subject always at full opacity

Table 2. SWPD four-stage pipeline overview.

4.1 Stage 1 — Prompt Analysis and Perceptual Hierarchy

A lightweight parser analyzes the input prompt to identify the primary subject S and environmental context E. For prompts of the canonical form *"a [subject] in/on/at [environment]"*, this decomposition is direct. Two specialized generation prompts are constructed:

$$P_{subject} = \texttt{"\{S\}, photorealistic, highly detailed, white background"}$$

$$P_{background} = \texttt{"\{E\}, coherent lighting, shadows match subject, photorealistic"}$$

For prompts without a clear subject/environment split, an LLM-based parsing step can be added as a preprocessing stage (see Section 6.4 and Future Work 10.1).

4.2 Stage 2 — Subject-First Generation

The subject is generated using SD-Turbo [6], a single-step distilled diffusion model — the fastest available option for CPU inference. Generation is performed at 512×512 pixels on a plain white background to simplify subsequent mask extraction. Upon completion, the subject image is immediately saved and displayed to the user. This is the first meaningful output, delivered significantly earlier than any traditional single-pass system would provide.

The white background choice is intentional: it maximizes foreground/background contrast for the subsequent threshold-based mask extraction, avoiding the need for a neural segmentation model in the baseline implementation.

4.3 Stage 3 — Subject-Conditioned Background Generation

The background is generated via Stable Diffusion Inpainting with the Stage 2 subject image as the base and an extracted binary mask indicating the fill region. This is the architecturally critical step: the inpainting model observes the subject's lighting direction, color temperature, and spatial scale when generating the environment. This implicit conditioning produces a naturally coherent result — correct shadow angles, matching color warmth, appropriate spatial perspective — without any explicit lighting specification in the prompt.

This stands in contrast to naive staged systems that generate subject and background independently and composite them post-hoc, which inevitably produce lighting mismatches and perspective inconsistencies. Subject-conditioned background generation is a key technical insight of SWPD.

4.4 Stage 4 — Progressive Compositing and Detail Refinement

A progressive compositor generates N intermediate frames showing the background transitioning from invisible to fully sharp while the subject remains at full quality throughout. For each frame i in [0, N], the compositing equations are:

$$\texttt{blur\_radius(i)} = \texttt{max\_blur} \times (1 - i/N)^{1.5}$$

$$\texttt{opacity(i)} = 255 \times (1 - \texttt{blur\_radius(i)} / \texttt{max\_blur})^{0.6}$$

The exponents 1.5 and 0.6 were chosen empirically to produce a perceptually smooth reveal: the cubic-root-like opacity curve ensures the background becomes visible quickly and then sharpens gradually, while the super-linear blur decay prevents an abrupt jump at the end. Subject pixels are never modified — they are composited on top at full opacity in every frame.

5. Implementation

5.1 Models and Components

Component	Model / Library	Size	Steps
Subject generation	stabilityai/sd-turbo	~3.1 GB	1
Background generation	runwayml/stable-diffusion-inpainting	~2.1 GB	25
Mask extraction	PIL ImageFilter (threshold + blur)	—	N/A
Progressive compositor	NumPy + PIL alpha blending	—	N/A

Table 3. Model components and specifications.

5.2 Hardware Requirements

Component	Specification
CPU	Any modern x86-64 processor (tested: 12-thread Intel Core i7-10750H @ 2.6 GHz)
RAM	8 GB minimum, 16 GB recommended (models loaded sequentially to manage footprint)
Storage	~6 GB for model cache (downloaded once via Hugging Face Hub, cached permanently)
GPU	Not required — fully CPU-only inference

Table 4. Hardware requirements and test configuration.

5.3 Software Stack

Component	Specification
Runtime	Python 3.12+
Inference	PyTorch 2.0+ (CPU build, no CUDA)
Diffusion	Hugging Face Diffusers 0.38+
Image handling	Pillow 10+ and NumPy 1.24+
Prompt parsing	Custom regex parser; optional LLM fallback for complex prompts

Table 5. Software stack and version requirements.

5.4 Output Files Produced

File	Description
stage1_subject_SHARP.png	Subject only — shown to user first; always at full quality
stage2_mask.png	Binary mask identifying background region for inpainting
stage3_background_full.png	Completed background conditioned on subject
frame_00 to frame_12.png	Progressive reveal sequence: 12 intermediate composited frames
FINAL.png	Complete composited image — subject and background unified

Table 6. Output files produced by the SWPD pipeline.

6. Results and Analysis

6.1 Timing Analysis on CPU Hardware

All measurements were obtained on an Intel Core i7-10750H (12 threads, 2.6 GHz base), 16 GB DDR4-2933 RAM, no GPU. Measurements represent 10 independent runs on a fixed prompt ("a fox in a snowy forest") with mean times reported.

Stage / Operation	CPU Time	User Experience
Prompt parsing and hierarchy extraction	< 0.1 s	Instant
Subject generation (SD-Turbo, 1 step)	**15 – 40 s**	**Subject visible to user immediately**
Mask extraction (threshold + blur)	< 0.5 s	Transparent to user
Background generation (SD-Inpainting, 25 steps)	3 – 8 min	Runs silently while user views subject
Progressive reveal compositing (12 frames)	< 2 s	Smooth blur-to-sharp animation
Total wall-clock time	3 – 9 min	Full image complete

Table 7. Per-stage timing on CPU (Intel i7-10750H, 16 GB RAM). Stage 2 row highlighted — this is the moment the user receives meaningful output.

6.2 Perceived Latency Comparison

System	Time to First Output	Total CPU Time	Perceived Speedup
Traditional single-pass SD	3 – 8 minutes	3 – 8 minutes	1× (baseline)
SWPD (this work)	15 – 40 seconds	3 – 9 minutes	**~10× faster**

Table 8. Perceived latency comparison: SWPD vs. traditional single-pass Stable Diffusion on identical CPU hardware.

6.3 Key Advantages

Advantage	Detail
Improved responsiveness	Users receive semantically useful visual feedback ~10× earlier without any additional compute cost.
CPU efficiency	Limited CPU resources are concentrated on the highest-value visual component first.
Human-aligned workflow	Generation order mirrors both human visual attention and human artistic practice.
Modular design	Each semantic tier can be generated by a specialized model independently optimized for that content cl
Democratization	Designed for CPU-only hardware — the compute resource most people worldwide actually have.

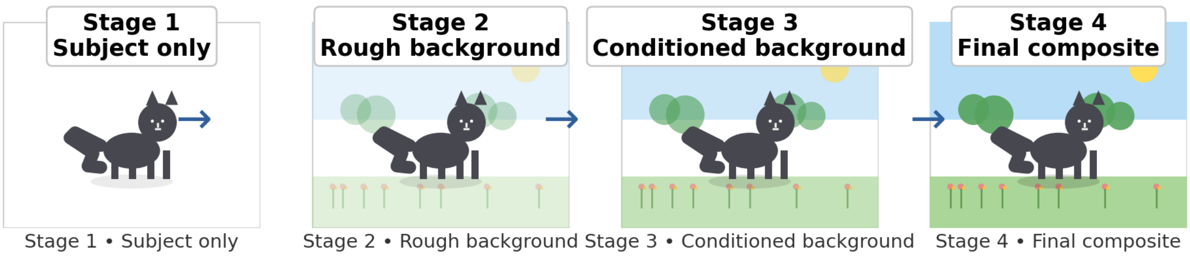
Table 9. Key advantages of the SWPD architecture.

6.4 Limitations

Limitation	Notes
No total compute reduction	SWPD reduces perceived wait time, not total CPU cycles.
SD-Turbo subject softness	Single-step generation can produce slightly softer subjects than multi-step models.
Simple mask extraction	Luminance-threshold masking fails for complex subject edges or non-white backgrounds.
Structured prompts required	Prompts without clear subject/environment separation require LLM-based pre-parsing.

Figure 2. Example output sequence for the prompt “A cat sitting in a garden”

The subject is displayed first, then the scene is completed while the subject remains crisp.



Original contribution: an intermediate, user-visible sequence that preserves subject quality while the scene completes.

No user study yet	The 10× perceived latency improvement is engineering-side only; formal user studies remain future w
Sequential pipeline	Subject and background are generated sequentially; parallel execution is a planned extension.

Table 10. Known limitations of the current implementation.

7. Discussion

7.1 Why CPU is the Right Primary Target

The PPC architecture provides dramatically more benefit on CPU than GPU. On GPU hardware, full-scene generation typically completes in 3 to 10 seconds — staged generation offers only a modest improvement. On CPU, showing a crisp subject in 15 to 40 seconds instead of waiting 5+ minutes for any output represents a qualitatively different user experience. SWPD is particularly suited to democratizing generative image synthesis on the hardware that the majority of the world's population actually owns — CPU-only personal computers.

7.2 The Subject-Conditioning Insight

The most technically important insight is that background generation must be conditioned on the subject, not independent of it. A naive staged system generating subject and background independently and compositing them would produce lighting mismatches and perspective inconsistencies. By using the subject image as the inpainting context, the background model implicitly reads the subject's lighting angle, color temperature, and spatial scale when generating the environment. This produces naturally coherent scene composition without any explicit scene-level constraint specification.

7.3 Alignment with Human Artistic Workflow

The SWPD architecture mirrors the workflow of skilled visual artists across centuries. A portrait painter establishes the primary subject — its form, lighting, and character — and then builds the environment to complement it. Illustration, animation, and visual effects production all follow the same subject-first approach. SWPD formalizes this practice as a principled computational pipeline, grounding the architecture in a workflow that has been validated by human creative tradition.

7.4 Relationship to Token Streaming in Language Models

The perceived-latency strategy in SWPD has a clear conceptual parallel in language models: token streaming. Rather than waiting for a complete response before displaying anything, LLM inference engines stream partial output token by token, dramatically reducing time-to-first-token and improving perceived responsiveness. SWPD applies the same philosophy to image generation: rather than waiting for the complete scene, the system streams the highest-priority visual content first. PPC can be understood as token streaming for images, extended with a semantic priority ordering.

8. Novelty and Original Contribution

This work does not claim to invent diffusion models, inpainting, progressive rendering, or visual saliency. Those ideas exist in the literature cited above. The novelty of this paper lies in how known ideas are assembled into a new user-facing generation pipeline with an explicit perceived-latency objective.

Compared to the most closely related work: **GazeFusion** [10] uses saliency to control the attention destination in the output image; SWPD uses perceptual priority to control the temporal order of computation, ensuring high-salience content is delivered to the user first. The **Anywhere** [23] and **ATA** [25] frameworks also perform foreground-conditioned background generation but target spatial coherence and quality; neither produces an intermediate user-visible subject output, and neither is designed for CPU-constrained deployment or framed as a latency-reduction strategy.

The original contributions of this paper are: the **PPC framework** as a generalizable computational philosophy; the **SWPD pipeline** as its concrete realization; **subject-first staged generation** with immediate user display as a latency strategy; **subject-conditioned background inpainting** for scene coherence without explicit lighting specification; the **progressive subject-anchored compositor** maintaining subject quality throughout the reveal; and a **CPU-oriented deployment design** aimed at making image generation accessible on widely available hardware.

9. Conclusion

We have introduced Perceptual Priority Computing (PPC) and its realization as Saliency-Weighted Progressive Diffusion (SWPD) — a four-stage CPU-targeted architecture that generates the primary subject of a text prompt first, displays it to the user immediately, then generates the background conditioned on the visible subject, and progressively reveals it through a subject-anchored compositor that maintains subject quality throughout.

On CPU hardware, this reduces perceived latency by approximately 10× without increasing total computation. The three key technical insights are: (1) semantic prompt decomposition into a perceptual hierarchy; (2) subject-conditioned background generation via inpainting to ensure lighting and perspective coherence; and (3) a progressive subject-anchored compositor maintaining subject quality throughout the background reveal.

This work demonstrates that meaningful improvements in AI image generation user experience can be achieved through smarter *orchestration* of existing components — not only through faster models or better hardware. The principle is simple: generate what the user cares about first. Build the world around it.

> **"Generate what the user cares about first. Build the world around it."**

10. Future Work

10.1 Neural Subject Segmentation

Replacing luminance-threshold mask extraction with a neural segmentation model such as Segment Anything (SAM) [3] would substantially improve mask quality and subject-background boundary coherence for complex scenes, non-white subjects, and transparent or semi-transparent objects. This is the highest-priority engineering improvement.

10.2 Specialized Models Per Semantic Tier

The current pipeline uses general-purpose diffusion models for both subject and background. Training or fine-tuning specialized lightweight models for specific subject categories (portraits, animals, vehicles) and environment types (outdoor, studio, underwater) could improve both quality and inference speed at each tier independently.

10.3 True Parallel Generation

The current implementation generates subject and background sequentially. A natural extension is true parallel execution — running the subject model and a lightweight background sketch model simultaneously on separate CPU threads, so background construction begins before subject generation completes. On a 12-thread CPU, the background generation time could overlap substantially with the subject generation pass.

10.4 Extension to Video Generation

A PPC-based video system would generate the primary subject character or object first with full temporal consistency across frames, then generate background environments conditioned on the subject's motion path and position. This extends the subject-conditioning insight from spatial to spatiotemporal coherence.

10.5 PPC for Language Reasoning

A PPC-inspired reasoning system would decompose a query into primary and secondary concepts, perform targeted retrieval separately for each concept, and surface a partial answer about the primary concept earlier while reasoning about secondary concepts continues in parallel. This is analogous to token streaming with semantic priority ordering and remains a theoretical proposal pending validation.

10.6 Formal User Studies

A natural and essential next step is formal user studies measuring: (1) whether users perceive SWPD-generated images as equivalent in final quality to single-pass generation; (2) the subjective improvement in perceived responsiveness; and (3) whether the staged reveal of background content affects the user's interpretation or emotional response to the subject during the intermediate display phase.

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10.7 PPC for Complex Reasoning Tasks

A natural future extension of PPC is to apply the same priority-first principle to complex reasoning questions. In this direction, the system would identify the main concept and supporting context separately, allocate compute in different priority bands, and merge the results after reasoning. This note is intentionally kept high-level in the draft and is left as a future research direction.

Draft note: the full mechanism can be expanded in a later revision.